

Exercise Sheet 8

Exercise 1 – Hybridization in Cuprates (CuO₂ layers)

We consider a square-planar CuO₂ unit: a central Cu²⁺ ion (with a hole in the 3d_{x²-y²} orbital) surrounded by four oxygen ions at positions:

$$\mathbf{R}_1 = (+a, 0), \quad \mathbf{R}_2 = (0, +a), \quad \mathbf{R}_3 = (-a, 0), \quad \mathbf{R}_4 = (0, -a),$$

in the xy -plane.

The relevant oxygen orbitals are the in-plane $2p$ orbitals pointing toward copper—the so-called $2p_\sigma$ orbitals:

$$\begin{aligned} |p_1\rangle &\equiv \text{O at } (+a, 0) : 2p_x \text{ (lobe points left, toward Cu)}, \\ |p_2\rangle &\equiv \text{O at } (0, +a) : 2p_y \text{ (lobe points down, toward Cu)}, \\ |p_3\rangle &\equiv \text{O at } (-a, 0) : -2p_x \text{ (lobe points right, toward Cu)}, \\ |p_4\rangle &\equiv \text{O at } (0, -a) : -2p_y \text{ (lobe points up, toward Cu)}. \end{aligned}$$

We adopt the convention that all orbitals point *toward* the Cu site, i.e. the sign is chosen so that the wavefunction has positive amplitude along the bond direction. Thus the four orthonormal oxygen orbitals (before hybridization) are:

$$|\psi_1\rangle = |p_x^{(+a,0)}\rangle, \quad |\psi_2\rangle = |p_y^{(0,+a)}\rangle, \quad |\psi_3\rangle = -|p_x^{(-a,0)}\rangle, \quad |\psi_4\rangle = -|p_y^{(0,-a)}\rangle.$$

The Cu 3d_{x²-y²} orbital has symmetry transforming as the irreducible representation B_{1g} of the D_{4h} point group: it changes sign under 90° rotation and is even under reflection in the xz/yz planes.

We now construct symmetry-adapted linear combinations (SALCs) of the four O p_σ orbitals. The character table of the C_{4v} subgroup (since the orbitals lie in-plane, and we only care about in-plane symmetries) tells us that the four basis functions span:

$$\Gamma = A_1 \oplus B_1 \oplus E.$$

We construct them explicitly.

- (i) Symmetric (totally symmetric, A_1) combination:

$$|\phi_{A_1}\rangle = \frac{1}{2} (|\psi_1\rangle + |\psi_2\rangle + |\psi_3\rangle + |\psi_4\rangle).$$

This is gerade, even under all symmetry operations—but note: the d_{x²-y²} orbital is antisymmetric under 90° rotation, so this combination has *zero* overlap with Cu d_{x²-y²}.

- (ii) B_1 symmetry (same as d_{x²-y²}): transforms as $x^2 - y^2$:

$$|\phi_{B_1}\rangle = \frac{1}{2} (|\psi_1\rangle - |\psi_2\rangle + |\psi_3\rangle - |\psi_4\rangle).$$

Check: Under 90° rotation ($x \rightarrow y, y \rightarrow -x$):

$$|\psi_1\rangle \rightarrow |\psi_2\rangle, \quad |\psi_2\rangle \rightarrow |\psi_3\rangle, \quad |\psi_3\rangle \rightarrow |\psi_4\rangle, \quad |\psi_4\rangle \rightarrow |\psi_1\rangle,$$

so

$$|\phi_{B_1}\rangle \rightarrow \frac{1}{2} (|\psi_2\rangle - |\psi_3\rangle + |\psi_4\rangle - |\psi_1\rangle) = -|\phi_{B_1}\rangle,$$

as expected for B_1 . This combination *does* have nonzero overlap with Cu 3d_{x²-y²}, and it is the bonding/antibonding partner in the Zhang–Rice picture.

- (iii) The remaining two combinations form a degenerate E representation (transforms like (x, y)). We can choose:

$$\begin{aligned} |\phi_{E,x}\rangle &= \frac{1}{\sqrt{2}} (|\psi_1\rangle - |\psi_3\rangle), \\ |\phi_{E,y}\rangle &= \frac{1}{\sqrt{2}} (|\psi_2\rangle - |\psi_4\rangle). \end{aligned}$$

But note: these are not yet orthogonal to $|\phi_{B_1}\rangle$? Let's check overlap:

$$\langle \phi_{B_1} | \phi_{E,x} \rangle = \frac{1}{2\sqrt{2}} (1 - 0 + 1 - 0) = \frac{1}{\sqrt{2}} \neq 0.$$

So better to use orthogonal SALCs.

Standard SALC construction for square geometry gives:

$$\begin{aligned} |\sigma_{a_1}\rangle &= \frac{1}{2}(|1\rangle + |2\rangle + |3\rangle + |4\rangle), \\ |\sigma_{b_1}\rangle &= \frac{1}{2}(|1\rangle - |2\rangle + |3\rangle - |4\rangle), \\ |\sigma_{b_2}\rangle &= \frac{1}{2}(|1\rangle + |2\rangle - |3\rangle - |4\rangle), \\ |\sigma_e\rangle &= \frac{1}{\sqrt{2}}(|1\rangle - |3\rangle), \quad \text{and} \quad \frac{1}{\sqrt{2}}(|2\rangle - |4\rangle). \end{aligned}$$

But B_2 transforms as xy , and E as (x, y) . However, in C_{4v} , the 4-dimensional representation reduces as $A_1 \oplus B_1 \oplus E$; there is no B_2 for this basis if orbitals are directed radially—because the B_2 SALC $(|1\rangle + |2\rangle - |3\rangle - |4\rangle)$ corresponds to d_{xy} -like symmetry, which cannot overlap with p_σ orbitals pointing radially (the overlap integral vanishes by symmetry).

Actually, careful: the four O p_σ orbitals pointing toward the center transform as the same representation as the vectors from center to corners, i.e. as the σ -bonding basis. Their irreducible decomposition is:

$$\Gamma = A_1 \oplus B_1 \oplus E.$$

Where:

- A_1 : s -like, totally symmetric,
- B_1 : $d_{x^2-y^2}$ -like,
- E : doubly degenerate, p_x/p_y -like.

The three combinations orthogonal to the B_1 one are:

$$\begin{aligned} |\chi_1\rangle &= \frac{1}{2}(|\psi_1\rangle + |\psi_2\rangle + |\psi_3\rangle + |\psi_4\rangle) & (A_1), \\ |\chi_2\rangle &= \frac{1}{2}(|\psi_1\rangle + |\psi_2\rangle - |\psi_3\rangle - |\psi_4\rangle) & (B_2\text{-like, but actually part of } E?), \\ |\chi_3\rangle &= \frac{1}{\sqrt{2}}(|\psi_1\rangle - |\psi_3\rangle) & \text{or alternative.} \end{aligned}$$

However, the correct orthonormal set (orthogonal to $|\phi_{B_1}\rangle$) is:

$$\begin{aligned} |\Sigma_{A_1}\rangle &= \frac{1}{2}(|\psi_1\rangle + |\psi_2\rangle + |\psi_3\rangle + |\psi_4\rangle), \\ |\Sigma_{B_2}\rangle &= \frac{1}{2}(|\psi_1\rangle - |\psi_2\rangle - |\psi_3\rangle + |\psi_4\rangle), \\ |\Sigma_E^{(1)}\rangle &= \frac{1}{\sqrt{2}}(|\psi_1\rangle - |\psi_3\rangle), \\ |\Sigma_E^{(2)}\rangle &= \frac{1}{\sqrt{2}}(|\psi_2\rangle - |\psi_4\rangle). \end{aligned}$$

But note: $\dim = 4$, so only three are orthogonal to the B_1 combination. The B_1 combination is:

$$|\Sigma_{B_1}\rangle = \frac{1}{2}(|\psi_1\rangle - |\psi_2\rangle + |\psi_3\rangle - |\psi_4\rangle).$$

One verifies orthonormality:

$$\langle \Sigma_i | \Sigma_j \rangle = \delta_{ij}, \quad \text{for } i, j \in \{A_1, B_1, B_2, E_x\}, \text{ etc.}$$

****Overlap with Cu $d_{x^2-y^2}$:** Only the B_1 combination has the same symmetry and thus nonzero overlap (by symmetry selection rules). The others have: - A_1 : symmetric under 90° rotation $\rightarrow$ orthogonal to B_1 , - B_2 : transforms as xy (odd under $x \leftrightarrow y$, even under 180°) $\rightarrow$ orthogonal, - E : transforms as (x, y) (vector-like) $\rightarrow$ also orthogonal.

Thus, the three orbitals orthogonal to the Cu $d_{x^2-y^2}$ symmetry are those of symmetries:

$$\boxed{A_1 \text{ (totally symmetric), } B_2 \text{ (} d_{xy}\text{-like), and } E \text{ (} p_x/p_y\text{-like, doubly degenerate)}}.$$

However, since we only need *three* orbitals orthogonal to the B_1 one (and the space is 4D), we can pick any orthonormal basis of the orthogonal complement. Conventionally, the nonbonding oxygen orbitals in cuprates are: - Two orbitals of e_g symmetry (in-plane p_x, p_y combinations forming E), - One orbital of a_{1g} symmetry (in-plane s -like, A_1).

But note: in the square-planar ligand field, the four O p_σ SALCs split as:

$$a_{1g} \oplus b_{1g} \oplus e_g,$$

where: - b_{1g} : bonds with Cu $d_{x^2-y^2}$ (forms bonding/antibonding), - a_{1g} : symmetric combination—nonbonding (no Cu s or d_{z^2} in minimal model), - e_g : two orbitals transforming as (x, y) —nonbonding (no Cu p orbitals in minimal Hubbard model).

Thus, the symmetries of the three orthogonal, non-overlapping orbitals are:

$$\boxed{a_{1g} \quad (\text{totally symmetric}), \quad e_g \quad (\text{doubly degenerate, } p_x/p_y\text{-like})}.$$

Since e_g is two-dimensional, it accounts for *two* orbitals; together with a_{1g} , that gives the required three.

Answer: The three orthogonal, non-overlapping orbitals have symmetries a_{1g} (one orbital) and e_g (two orbitals, degenerate). In D_{4h} notation: A_{1g} and E_g ; or in C_{4v} : A_1 and E .

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Exercise 2 – The sp^2 hybridized orbitals in graphite

We are to construct three orthonormal sp^2 hybrid orbitals for carbon in the plane, 120° apart. Let the carbon atom sit at the origin. The hybrids point toward the three nearest neighbors in the honeycomb lattice, i.e. along directions:

$$\mathbf{n}_1 = (1, 0), \quad \mathbf{n}_2 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad \mathbf{n}_3 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

An sp^2 hybrid orbital pointing in direction $\mathbf{n} = (\cos \theta, \sin \theta, 0)$ is a linear combination:

$$|h(\theta)\rangle = a|2s\rangle + b(\cos \theta |2p_x\rangle + \sin \theta |2p_y\rangle).$$

By symmetry, the coefficients a, b are the same for all three hybrids.

We require:

- (a) Normalization: $\langle h(\theta)|h(\theta)\rangle = 1$,
- (b) Orthogonality: $\langle h(\theta_i)|h(\theta_j)\rangle = 0$ for $i \neq j$,
- (c) The three hybrids lie in the xy -plane and sum to zero (closure).

Assume normalized atomic orbitals:

$$\langle 2s|2s\rangle = 1, \quad \langle 2p_x|2p_x\rangle = 1, \quad \langle 2p_y|2p_y\rangle = 1, \quad \text{all cross terms} = 0.$$

Step 1: Ansatz

Let:

$$\begin{aligned} |h_1\rangle &= a|2s\rangle + b|2p_x\rangle, \quad (\theta = 0^\circ) \\ |h_2\rangle &= a|2s\rangle + b\left(-\frac{1}{2}|2p_x\rangle + \frac{\sqrt{3}}{2}|2p_y\rangle\right), \quad (\theta = 120^\circ) \\ |h_3\rangle &= a|2s\rangle + b\left(-\frac{1}{2}|2p_x\rangle - \frac{\sqrt{3}}{2}|2p_y\rangle\right), \quad (\theta = 240^\circ) \end{aligned}$$

Step 2: Orthogonality condition

Compute $\langle h_1|h_2\rangle$:

$$\langle h_1|h_2\rangle = a^2 + b^2 \left(1 \cdot \left(-\frac{1}{2}\right)\right) = a^2 - \frac{1}{2}b^2.$$

Set to zero:

$$a^2 = \frac{1}{2}b^2.$$

Step 3: Normalization

$$\langle h_1 | h_1 \rangle = a^2 + b^2 = 1.$$

Substitute $a^2 = \frac{1}{2}b^2$:

$$\frac{1}{2}b^2 + b^2 = \frac{3}{2}b^2 = 1 \quad \Rightarrow \quad b^2 = \frac{2}{3}, \quad a^2 = \frac{1}{3}.$$

Choose real, positive coefficients (phase convention):

$$a = \frac{1}{\sqrt{3}}, \quad b = \sqrt{\frac{2}{3}}.$$

Step 4: Final orthonormal sp^2 hybrids

$$\begin{aligned} |h_1\rangle &= \frac{1}{\sqrt{3}}|2s\rangle + \sqrt{\frac{2}{3}}|2p_x\rangle, \\ |h_2\rangle &= \frac{1}{\sqrt{3}}|2s\rangle + \sqrt{\frac{2}{3}}\left(-\frac{1}{2}|2p_x\rangle + \frac{\sqrt{3}}{2}|2p_y\rangle\right), \\ |h_3\rangle &= \frac{1}{\sqrt{3}}|2s\rangle + \sqrt{\frac{2}{3}}\left(-\frac{1}{2}|2p_x\rangle - \frac{\sqrt{3}}{2}|2p_y\rangle\right). \end{aligned}$$

Verification: - Angles: direction vectors are $\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3$ — 120° apart. - Orthonormal:

$$\langle h_1 | h_2 \rangle = \frac{1}{3} - \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3} - \frac{1}{3} = 0,$$

and similarly for others. Normalization:

$$\langle h_1 | h_1 \rangle = \frac{1}{3} + \frac{2}{3} = 1.$$

- Sum to zero:

$$|h_1\rangle + |h_2\rangle + |h_3\rangle = \sqrt{3}|2s\rangle + \sqrt{\frac{2}{3}}\left(1 - \frac{1}{2} - \frac{1}{2}\right)|2p_x\rangle + \sqrt{\frac{2}{3}}\left(0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}\right)|2p_y\rangle = \sqrt{3}|2s\rangle,$$

Wait — not zero! But note: the *vector directions* sum to zero, but the *orbital coefficients* do not annihilate the s -part. However, physical hybrids need not sum to zero; rather, the three hybrids use 1 s and 2 p electrons, leaving one p_z orbital unhybridized (for π -bonding in graphite). Indeed:

$$|h_1\rangle + |h_2\rangle + |h_3\rangle = \sqrt{3}|2s\rangle,$$

which is fine — the s -orbital contributes equally to all three.

Thus, the above are the correct orthonormal sp^2 hybrids.

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Exercise 3 – Different bonding types

We discuss the dominant bonding type(s) for each material.

- **Krypton (Kr):** Noble gas; solid at low temperatures. Binding is via **van der Waals (dispersion) forces**. No permanent dipole — purely induced dipole-induced dipole (London dispersion).
Dominant: van der Waals.
- **Table salt (NaCl):** Classic ionic crystal. Electrons transferred from Na to Cl. Strong Coulomb attraction. Small covalent contribution ($\sim 10\%$), but negligible for classification.
Dominant: ionic.
- **Sodium (Na):** Alkali metal. Delocalized valence electrons in a sea of positive ions. Metallic bonding.
Dominant: metallic.

- **Graphite:** Carbon layers. *Intralayer:* each C is sp^2 hybridized — strong directional σ bonds (covalent), plus delocalized π electrons (partially metallic/conjugated). *Inter-layer:* weak van der Waals. Overall, in-plane covalent dominates mechanical/electronic properties.
Dominant: covalent (with metallic character in-plane, van der Waals inter-layer).
- **Diamond:** Carbon, sp^3 hybridized, tetrahedral network. Pure directional covalent bonds.
Dominant: covalent.
- **Argon (Ar):** Like krypton — noble gas solid.
Dominant: van der Waals.
- **Gallium arsenide (GaAs):** III–V semiconductor. Significant ionicity (Pauling electronegativity: Ga 1.81, As 2.18 $\rightarrow \Delta\chi = 0.37$, ionic character $\sim 4\%$), but bonding is primarily **covalent with partial ionic character** (polar covalent). Zinc-blende structure resembles diamond.
Dominant: covalent (polar covalent).
- **Zinc oxide (ZnO):** II–VI semiconductor. Larger electronegativity difference (Zn 1.65, O 3.44 $\rightarrow \Delta\chi = 1.79$, ionic character $\sim 55\%$). Wurtzite structure. Bonding is **mixed ionic–covalent**, often considered predominantly ionic, but with strong covalent contribution due to orbital hybridization. Literature often calls it “partially ionic”.
Dominant: ionic–covalent (ionic character significant, but covalent contribution non-negligible).
- **Quartz (SiO₂):** Network solid. Each Si is tetrahedrally coordinated by O; each O bridges two Si. Pauling electronegativity: Si 1.90, O 3.44 $\rightarrow \Delta\chi = 1.54 \rightarrow$ ionic character $\sim 45\%$. However, the directional Si–O–Si bonds and high melting point suggest strong **covalent character within a polar network**. Often described as a **covalent network solid with ionic contributions**.
Dominant: covalent (polar covalent network).
- **Ammonia (NH₃):** Molecular solid (below 195 K). Molecules held by **hydrogen bonds** (a strong type of dipole–dipole interaction), plus weaker van der Waals. Intramolecular N–H bonds are polar covalent. In solid, intermolecular H-bonding dominates cohesion.
Dominant: hydrogen bonding (a subclass of dipole–dipole/van der Waals, but distinct in strength and directionality).
- **Tetrafluoromethane (CF₄):** Symmetric tetrahedral molecule. C–F bonds are polar covalent, but molecule is nonpolar (T_d symmetry, dipole moment = 0). Solid is held by **van der Waals (dispersion) forces**.
Dominant: van der Waals.

Summary table:

Material	Dominant bonding type
Krypton	van der Waals
NaCl	ionic
Sodium	metallic
Graphite	covalent (in-plane) + van der Waals (interlayer)
Diamond	covalent
Argon	van der Waals
GaAs	polar covalent
ZnO	ionic–covalent (moderately ionic)
SiO ₂ (quartz)	polar covalent network
NH ₃	hydrogen bonding
CF ₄	van der Waals

Note: Real materials often exhibit mixed bonding; the classification refers to the *dominant* contribution to cohesion.